

**Program: Bachelor of Engineering in Computer Science & Engineering**

*A Project Report on*

**Tactic Sense**

*Submitted in partial fulfillment of the requirements for the course*

***CSP67: MINI PROJECT***

*By*

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**CERTIFICATE**

Certified that the project work titled “Tactic Sense” carried out by Akanksh N(1MS23CS018), Tanvi Purugulla(1MS23CS143), Yuvan Jayakumar (1MS23CS225) are bonafide students of M. S. Ramaiah Institute of Technology, Bengaluru, in partial fulfillment of the course Mini Project CSP67 during the term March-June 2025. The project report has been approved as it satisfies the academic requirements for the aforesaid course. To the best of our understanding, the work submitted in this report is the original work of students.

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- 2.

# DECLARATION

We, hereby, declare that the entire work embodied in this mini project report has been carried out by us at M. S. Ramaiah Institute of Technology, Bengaluru, under the supervision of , **Dr Akshata S Bhayyar, Assistant Professor**, Department of CSE. This report has not been submitted in part or full for the award of any diploma or degree of this or to any other university.

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## ABSTRACT

Negotiation and communication play an important role in modern business environments, customer interactions, online marketplaces, and financial discussions. During conversations, individuals often use manipulation tactics such as false urgency, emotional pressure, proof bombing, and authority influence to persuade others into making rushed decisions. Most existing systems focus mainly on sentiment analysis or speech transcription and fail to identify specific manipulation tactics during live conversations. In addition, traditional natural language processing systems struggle with multilingual and code-switched conversations commonly found in real-world communication environments. To address these limitations, the proposed project “TacticSense” was developed as a real-time negotiation tactic detection system capable of analyzing live conversations and identifying manipulative communication patterns instantly.

The project combines speech processing, multilingual natural language processing, contextual reasoning, and dashboard analytics into a unified conversational intelligence framework. The system captures live audio streams and converts speech into text using the Whisper speech recognition model. Speaker diarization is performed using Pyannote.audio to identify multiple speakers within a conversation. The generated transcripts are processed using XLM-RoBERTa for multilingual contextual embedding generation. A hierarchical transformer architecture analyzes conversational flow across multiple turns and performs multi-label classification to detect tactics such as false urgency, proof bombing, emotional pressure, and authority manipulation. The system additionally uses Redis caching and MongoDB storage for real-time processing and conversational analytics. Dashboard interfaces provide confidence scores, transcript annotations, heatmaps, and strategic counter-suggestions for users.

The experimental results demonstrate that transformer-based conversational intelligence systems significantly improve contextual understanding and manipulation tactic detection compared to traditional sentiment analysis approaches. The system successfully generated real-time alerts, confidence scores, and conversational analytics during testing.

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# Chapter 1

## INTRODUCTION

### 1.1 General Introduction

Communication and negotiation play an important role in business, online marketplaces, customer support systems, and financial discussions. During conversations, people often use manipulation tactics such as false urgency, emotional pressure, proof bombing, and authority influence to persuade others into making decisions quickly. These tactics are usually subtle and difficult to identify during live interactions. As a result, users often realize they were manipulated only after the conversation ends, which may lead to poor decision-making, financial loss, or emotional stress.

With the rapid growth of online meetings, virtual communication platforms, and digital negotiations, the need for intelligent conversational analysis systems has increased significantly. Existing systems mainly focus on sentiment analysis or simple speech transcription and are unable to detect specific negotiation tactics in real time. Traditional NLP models also struggle with multilingual and code-switched conversations, especially in Indian communication environments where English is mixed with regional languages.

The proposed project, TacticSense, is an AI-powered real-time negotiation tactic detection system designed to solve this problem. The system captures live conversations, converts speech into text, identifies speakers, maintains conversational context, and detects manipulation tactics using transformer-based deep learning models. The system also provides real-time alerts and counter-suggestions to help users respond strategically during negotiations.

TacticSense combines technologies such as Whisper speech recognition, XLM-RoBERTa embeddings, hierarchical transformers, Redis caching, and MongoDB analytics into a unified conversational intelligence framework.

### 1.2 Problem Statement

Manipulation tactics are increasingly being used in modern business negotiations, online sales discussions, customer interactions, and financial conversations. Techniques such as false urgency, emotional pressure, proof bombing, and persuasive framing often force users



into making rushed decisions without proper analysis. Since these tactics are hidden within natural conversations, many users fail to recognize them during live communication. Existing systems mainly focus on sentiment analysis, chatbot responses, or static text classification instead of identifying specific negotiation tactics in real time. Traditional natural language processing models also lack contextual reasoning and struggle to understand multi-turn conversational flow. Most systems additionally fail when handling multilingual and code-switched conversations such as Hinglish or mixed-language communication.

There is currently no commonly available system capable of:

- Detecting manipulation tactics in real time
- Understanding conversational context
- Supporting multilingual communication
- Providing counter-strategies to users
- Performing speaker-aware conversational analysis

The proposed TacticSense system addresses these challenges using speech recognition, multilingual transformers, hierarchical conversational analysis, and real-time dashboard analytics. The system continuously processes live audio conversations, identifies speakers, maintains contextual history, and detects negotiation tactics instantly.

The final output includes tactic alerts, confidence scores, annotated transcripts, and strategic counter-suggestions. By solving this problem, TacticSense improves transparency and helps users make informed decisions during live negotiations.

### **1.3 Objectives of the Project**

The main objective of TacticSense is to develop an AI-powered conversational intelligence system capable of detecting negotiation manipulation tactics during live conversations. The project combines speech processing, multilingual natural language understanding, and transformer-based contextual analysis into a single integrated framework.

The major objectives of the project are:

- To capture and process live conversational audio streams
- To convert speech into text using Whisper
- To identify speakers using speaker diarization
- To maintain conversational context using sliding windows

- To generate multilingual embeddings using XLM-RoBERTa
- To analyze conversation flow using hierarchical transformers
- To detect multiple negotiation tactics simultaneously
- To provide real-time alerts and confidence scores
- To generate strategic counter-suggestions for users
- To store conversational analytics and transcripts

Another important objective of the project is to support multilingual and code-switched communication commonly found in Indian conversations. The system is designed to understand mixed-language speech and contextual patterns more effectively than traditional NLP systems.

The project also demonstrates the practical application of transformer-based architectures in conversational intelligence systems. Through this implementation, users can better understand how AI can improve decision-making and reduce manipulation risks during negotiations.

## **1.4 Project Deliverables**

The TacticSense project produces multiple technical and functional deliverables related to conversational intelligence and real-time negotiation analysis. These deliverables demonstrate the successful integration of speech recognition, transformer models, dashboard analytics, and contextual conversational reasoning.

The primary deliverables of the project include:

1. Real-Time Negotiation Tactic Detection System, A live conversational analysis framework capable of identifying manipulation tactics such as false urgency, proof bombing, and emotional pressure.
2. Speech Processing Pipeline, Integration of Whisper speech recognition and Pyannote speaker diarization for processing live audio streams.
3. Hierarchical Transformer Architecture, A transformer-based conversational reasoning system capable of analyzing multi-turn conversational flow.

4. Dashboard Analytics System, Real-time dashboard alerts, tactic confidence scores, manipulation heatmaps, and annotated transcripts.
5. Database and Storage Framework, MongoDB and Redis integration for transcript storage, analytics, session management, and caching.
6. Performance Evaluation Metrics, Accuracy-loss graphs, precision-recall curves, confusion matrices, and model evaluation outputs.
7. Technical Documentation, Complete architecture diagrams, ER diagrams, methodology explanations, implementation details, and research documentation.

These deliverables collectively demonstrate the creation of a modern AI-powered conversational intelligence platform suitable for negotiation environments and real-time decision support systems.

## **1.5 Current Scope**

The current scope of TacticSense mainly focuses on detecting manipulation tactics during live conversational interactions such as business negotiations, customer support calls, online sales discussions, and professional communication environments. The system currently supports multilingual conversational analysis with primary emphasis on English-dominant and code-switched communication.

The implemented system includes:

- Live audio capture
- Speech-to-text conversion
- Speaker diarization
- Sliding window context analysis
- XLM-RoBERTa embeddings
- Hierarchical transformer reasoning
- Multi-label tactic classification
- Dashboard analytics and alerts

The current version focuses on detecting selected manipulation tactics including:

- False urgency
- Proof bombing
- Emotional pressure
- Authority manipulation

The project mainly functions as a prototype research framework developed for academic analysis and conversational intelligence experimentation. Although the architecture supports scalability, the present implementation primarily emphasizes contextual conversational understanding and real-time AI inference.

The project also demonstrates how transformer-based models can analyze conversational sequences instead of isolated sentences. This contextual understanding significantly improves negotiation tactic detection compared to traditional sentiment analysis systems.

## 1.6 Future Scope

Although TacticSense provides a strong conversational intelligence framework, several future improvements can significantly enhance the system's scalability, accuracy, and real-world applicability.

Future enhancements include:

1. **Video Meeting Integration**, The system can be integrated with platforms such as Zoom, Google Meet, and Microsoft Teams for live meeting analysis.
2. **Mobile Application Support**, A mobile-based implementation can provide real-time alerts during phone conversations.
3. **Emotion-Aware AI Systems**, Future models can combine facial emotion detection and vocal tone analysis with conversational NLP.
4. **Larger Multilingual Datasets**, Training on larger multilingual negotiation datasets can improve model generalization and accuracy.
5. **Lightweight Transformer Models**, Optimized transformer architectures can reduce computational requirements and improve deployment efficiency.
6. **Enterprise Analytics Platforms**, Organizations can use enterprise dashboards for communication monitoring and negotiation training.
7. **Browser Extension Integration**, The system can be extended as a browser plugin for online negotiation platforms.

These future improvements would make TacticSense more scalable, accurate, and commercially deployable across industries such as finance, customer support, business consulting, and communication intelligence.

## **Chapter 2**

### **PROJECT ORGANIZATION**

#### **2.1 Software Process Model**

The development of TacticSense follows an incremental and modular software development process. Since the project involves multiple technologies such as speech recognition, natural language processing, dashboard analytics, database management, and transformer-based deep learning models, the project was developed in separate stages for better flexibility and testing.

The first stage involved requirement analysis and understanding the problem of manipulation tactics in negotiations. Research was conducted on conversational AI systems, transformer architectures, speech processing frameworks, and multilingual NLP techniques. Based on this analysis, the overall system architecture and workflow were planned.

The second stage focused on system design and architecture development. Pipeline flow diagrams, ER diagrams, data flow structures, and conversational processing models were prepared. Technologies such as Whisper, XLM-RoBERTa, PyTorch, FastAPI, Redis, and MongoDB were selected based on system requirements.

The third stage involved implementation and integration of AI components. Speech recognition, speaker diarization, contextual embedding generation, and transformer models were integrated into the conversational pipeline. Backend APIs and dashboard systems were also implemented during this phase.

The fourth stage focused on testing and evaluation. Accuracy-loss graphs, confusion matrices, and precision-recall analysis were generated to evaluate model performance. The final stage included documentation, result analysis, and report preparation.

The modular software development process improved debugging, experimentation, scalability, and integration of multiple AI components within the TacticSense framework.

## 2.2 Roles and Responsibilities

The development of TacticSense required collaboration between multiple team members with responsibilities divided according to technical expertise and implementation requirements. The project included tasks related to frontend development, backend integration, AI model implementation, database management, testing, and documentation. The below table 2.1 it highlights the roles and responsibilities.

Table 2.1 Roles of team members

| Member                                 | Role                                           | Key Responsibilities                                                                                                                                                                                                                                                                                                                                                                                   | Primary Focus Areas                                                                                                                                                                                                                                            |
|----------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Akanksh</b><br>(1MS23CS018)         | <b>Project Lead &amp; Backend Developer</b>    | <ul style="list-style-type: none"><li>Leading overall project coordination</li><li>Designing system architecture and backend workflow</li><li>Implementing FastAPI services and WebSocket communication</li><li>Integrating frontend with AI/ML modules</li><li>Managing deployment and testing pipelines</li><li>Preparing documentation and reports</li></ul>                                        | <ul style="list-style-type: none"><li>Backend development using FastAPI</li><li>Real-time data streaming</li><li>System integration</li><li>API design and routing</li><li>Project planning and coordination</li><li>Testing and debugging</li></ul>           |
| <b>Tanvi P</b><br>(1MS23CS143)         | <b>AI / NLP Engineer</b>                       | <ul style="list-style-type: none"><li>Implementing speech recognition using Whisper model</li><li>Integrating XLM-RoBERTa for semantic understanding</li><li>Designing hierarchical transformer architecture</li><li>Developing multi-label tactic classification module</li><li>Building manipulation and pressure detection mechanisms</li><li>Model training, optimization and evaluation</li></ul> | <ul style="list-style-type: none"><li>AI model integration</li><li>NLP pipeline development</li><li>Embedding generation</li><li>Tactic detection algorithms</li><li>Model training and fine-tuning</li><li>Performance evaluation</li></ul>                   |
| <b>Yuvan Jayakumar</b><br>(1MS23CS225) | <b>Data Engineer &amp; Analytics Developer</b> | <ul style="list-style-type: none"><li>Managing MongoDB and Redis databases</li><li>Handling transcript storage and session data</li><li>Implementing analytics and heatmap generation</li><li>Data preprocessing and pipeline support</li><li>Result visualization and report export</li><li>Testing, debugging and documentation</li></ul>                                                            | <ul style="list-style-type: none"><li>Database design and management</li><li>Data storage and retrieval</li><li>Analytics and visualization</li><li>Heatmap and dashboard data</li><li>Testing and quality assurance</li><li>Technical documentation</li></ul> |

## **Chapter 3**

### **LITERATURE SURVEY**

#### **3.1 Introduction**

Literature survey plays an important role in understanding existing technologies, research methodologies, and limitations related to conversational intelligence and negotiation analysis systems. The primary goal of this survey is to study existing work in speech recognition, multilingual natural language processing, transformer architectures, sentiment analysis, and conversational AI systems relevant to negotiation tactic detection.

Modern conversational AI systems mainly focus on tasks such as speech transcription, chatbot generation, sentiment analysis, and voice assistants. Although these systems perform well for general communication tasks, they are not specifically designed to detect manipulation tactics during live conversations. Most traditional systems analyze isolated sentences rather than understanding the complete conversational flow across multiple turns.

Recent advancements in transformer architectures such as BERT, RoBERTa, and XLM-RoBERTa have significantly improved contextual understanding in NLP applications. Similarly, speech processing frameworks such as Whisper and Pyannote.audio have improved speech recognition and speaker diarization accuracy. These advancements make it possible to design systems capable of real-time conversational intelligence.

However, several research gaps still exist:

- Limited real-time negotiation tactic detection systems
- Poor multilingual conversational support
- Lack of speaker-aware contextual analysis
- Limited use of hierarchical conversational transformers
- Insufficient counter-strategy recommendation systems

The TacticSense project attempts to bridge these research gaps by integrating speech recognition, multilingual embeddings, contextual transformers, dashboard analytics, and real-time manipulation detection into a single framework. The figure 3.1 aims at showing the research workflow of the project.



**Research Workflow Diagram — Milestone Gate Style**

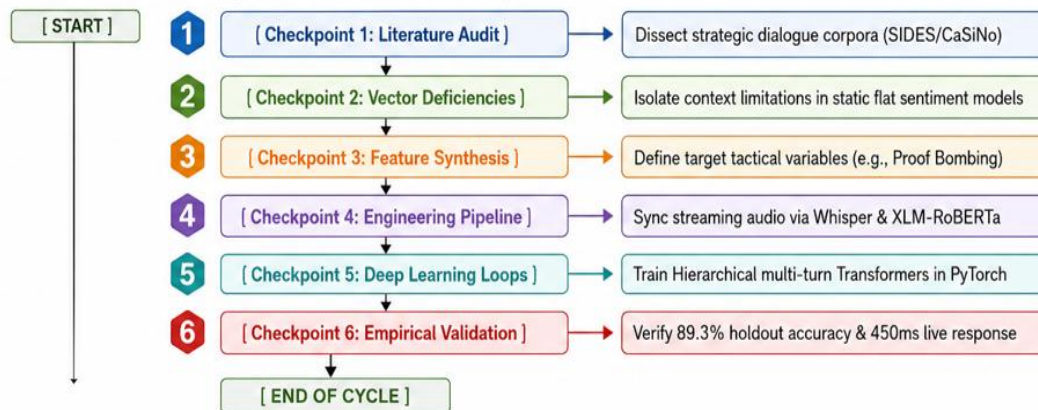


Fig 3.1-Research workflow diagram

### 3.2 Related Works with Citation of References

Several research studies and conversational AI systems have contributed toward speech processing, sentiment analysis, and contextual NLP. Existing work in transformer-based NLP has provided a strong foundation for building intelligent conversational systems.

The introduction of BERT by Google significantly improved contextual language understanding using bidirectional transformers. Later, RoBERTa improved BERT's training methodology and achieved higher performance in text classification tasks. However, these models mainly focused on English-language applications.

XLM-RoBERTa extended multilingual support and improved cross-lingual understanding for multilingual datasets. This model is highly useful in conversational environments involving code-switching and mixed-language communication, which is common in Indian negotiation scenarios.

OpenAI's Whisper speech recognition model demonstrated strong performance in multilingual speech-to-text conversion. Whisper supports noisy audio environments and improves transcription quality for real-world conversations. Similarly, Pyannote.audio introduced robust speaker diarization frameworks capable of identifying different speakers within audio streams.

Several studies also explored hierarchical transformers for dialogue understanding and conversational reasoning. These architectures analyze multi-turn conversations instead of isolated sentences, making them suitable for negotiation tactic detection.

Despite these advancements, existing systems still lack integrated real-time frameworks capable of:

- Detecting manipulation tactics
- Providing live alerts
- Supporting multilingual negotiation analysis
- Generating strategic counter-suggestions

TacticSense combines these technologies into a unified conversational intelligence pipeline specifically designed for negotiation environments. The below table 3.2 shows the research gap involved and how the Tactic sense solves it.

|  AUTHORS / SOURCE                 |  DATASET & FRAMEWORK                          |  METHODOLOGY USED                                                                                   |  CORE LIMITATIONS                                                                                                                                                                                                          |  HOW TACTICSENSE RESOLVES IT (THE GAP CLOSURE)                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Lewis et al. (Facebook Research) |  Deal or No Deal (End-to-End Negotiator)      |  End-to-end Supervised Learning (SL) and Reinforcement Learning (RL) on tokenized text sequences.   |  <b>Flat Context / Single-Issue:</b><br>Restricted to simplistic, short text bargaining for static items (books, hats). Completely ignores conversational multi-turn context arcs and live multi-label execution.          |  <b>Hierarchical Sequential Tracking:</b><br>Replaces end-to-end token generation with a multi-turn Hierarchical Transformer that detects high-level strategic patterns over extended live dialogues.                    |
|  Chawla et al. (USC / CaSiNo)     |  CaSiNo Dataset (Camp Site Negotiation)       |  Multi-task NLP classification models tracking self-interest vs. cooperative persuasion strategies. |  <b>Text-Only / Crowdsourced:</b><br>Relies entirely on static, asynchronous chat logs without acoustic parameters. Completely fails to address live verbal dynamics, real-time audio latency, or regional code-switching. |  <b>Real-Time Acoustic Pipeline:</b><br>Integrates a true live streaming pipeline (Whisper-large-v3 + Pyannote.audio) to process actual spoken conversations instead of offline text-only chat entries.                  |
|  Zhou et al. (Amazon-SIDES)     |  SIDES Corpus (Persuasion For Good)         |  Multi-turn text sequence encoders calculating singular macro-sentiment intent classes.           |  <b>Single-Label Focus:</b><br>Employs mutual exclusivity assumptions (Softmax classification) where an utterance can only map to one tag, completely missing blended, overlapping deceptive patterns.                   |  <b>Multi-Label Sigmoid Matrix:</b><br>Uses a multi-label classification head powered by independent Sigmoid activations to detect False Urgency and Proof Bombing occurring simultaneously in the same sentence.      |
|  Traditional NLP / Baseline     |  Standard Static Word Embedding Classifiers |  Flat text sequence encoding (TF-IDF / Basic BERT) evaluating sentences in isolation.             |  <b>Vocabulary Shock &amp; Zero Memory:</b><br>Out-of-vocabulary exceptions hit instantly when encountering regional Indian code-switching (Hinglish) and completely lacks speaker-to-tactic orientation.                |  <b>Cross-Lingual Turn Fusion:</b><br>Implements an out-of-the-box xlm-roberta-base backbone (250k token space) element-wise aggregated with learnable speaker embedding tensors to explicitly preserve who said what. |

Fig 3.2 -Research gap analysis table

### 3.3 Conclusion of Survey

The literature survey shows that modern AI systems have significantly improved speech recognition, contextual language understanding, and conversational processing. Transformer architectures such as BERT, RoBERTa, and XLM-RoBERTa have improved contextual reasoning capabilities, while Whisper and Pyannote.audio have enhanced speech transcription and speaker diarization performance.

Although these technologies provide strong foundations for conversational AI, most existing systems are limited to sentiment analysis, chatbot interactions, or speech transcription tasks. Very few systems focus specifically on real-time negotiation tactic detection and conversational manipulation analysis.

Existing research also reveals several important limitations:

- Limited multilingual conversational support
- Lack of real-time contextual reasoning
- Poor handling of code-switched conversations
- Absence of speaker-aware tactic analysis
- Limited integration of conversational analytics dashboards

The survey further indicates that hierarchical transformers provide better multi-turn conversational understanding compared to traditional NLP models. These architectures are capable of understanding conversational progression, emotional context, and persuasion patterns across multiple dialogue turns.

Based on the analysis of existing systems and research papers, TacticSense was designed as a unified AI-powered conversational intelligence framework. The project integrates multilingual embeddings, speech recognition, hierarchical transformers, dashboard analytics, and tactic detection into a single real-time system.

The literature survey therefore validates the importance and relevance of the proposed research work and highlights the need for intelligent negotiation tactic detection systems in modern communication environments.

## Chapter 4

### PROJECT MANAGEMENT PLAN

#### 4.1 Schedule of the Project

The development of TacticSense was planned systematically by dividing the project into multiple phases such as requirement analysis, research, architecture design, AI model integration, dashboard development, testing, and documentation. A structured schedule helped in organizing tasks efficiently and ensured timely completion of the project.

The initial phase focused on understanding negotiation manipulation tactics and studying existing conversational AI systems. Research was conducted on speech recognition, multilingual transformers, contextual NLP, and conversational intelligence architectures. Based on this analysis, the project objectives and workflow were finalized.

The second phase involved system design and preparation of architecture diagrams, ER diagrams, pipeline structures, and database schemas. Technologies such as Whisper, XLM-RoBERTa, PyTorch, FastAPI, Redis, and MongoDB were selected during this stage.

The third phase focused on implementation and integration of AI modules. Speech-to-text conversion, speaker diarization, transformer embeddings, and hierarchical conversational analysis were integrated into the processing pipeline. Dashboard interfaces and backend APIs were also developed during this phase.

The fourth phase involved testing, debugging, result analysis, and graph generation. Accuracy-loss graphs, confusion matrices, and precision-recall curves were generated to evaluate system performance.

The final phase included report preparation, PPT creation, screenshot collection, and final project documentation. The figure 4.1 shows the project plan in gantt chart.

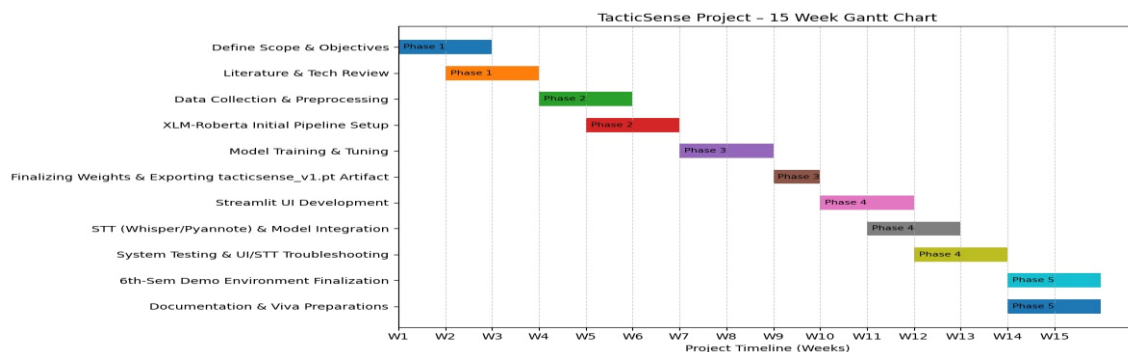


Fig 4.1-Gantt chart

## 4.2 Risk Identification

The development of TacticSense involves several technical and implementation-related risks due to the integration of multiple AI technologies and real-time conversational analysis systems. Identifying these risks early helps improve system reliability and project planning.

One major risk is speech recognition inaccuracy. Variations in accents, pronunciation, background noise, and overlapping speech may reduce transcription quality. Since the system depends heavily on speech-to-text conversion, incorrect transcripts can affect downstream NLP analysis.

Another important risk involves multilingual and code-switched conversations. Users may mix English with regional languages such as Hindi or Kannada, which can reduce contextual understanding if the model is not properly trained on multilingual data.

Real-time latency is another challenge. Hierarchical transformers and multilingual models require high computational power and memory. Delays in inference may affect the responsiveness of the dashboard alerts.

Dataset availability is also a significant concern. High-quality negotiation datasets with tactic annotations are limited. The project partially addresses this issue by combining public datasets with custom-generated conversational samples.

Additional risks include:

- Model overfitting
- Limited GPU resources
- API integration failures
- Database synchronization issues
- Scalability challenges during deployment

To reduce these risks, modular development, caching mechanisms, multilingual models, incremental testing, and optimized transformer architectures were used throughout the project.

## Chapter 5

# SOFTWARE REQUIREMENT SPECIFICATIONS

### 5.1 Purpose

The purpose of TacticSense is to develop a real-time conversational intelligence system capable of detecting manipulation tactics during live negotiations and discussions. The system analyzes conversations using speech processing and transformer-based NLP models to identify tactics such as false urgency, proof bombing, emotional pressure, and authority manipulation.

The project aims to assist users in recognizing manipulative communication before making important decisions. The system also provides confidence scores, annotated transcripts, dashboard analytics, and strategic counter-suggestions during conversations.

TacticSense demonstrates the practical use of conversational AI, multilingual transformers, and hierarchical contextual analysis for negotiation environments.

### 5.2 Project Scope

The scope of TacticSense focuses on real-time conversational analysis and negotiation tactic detection using AI-powered speech and NLP technologies. The project is primarily designed for business negotiations, online discussions, customer interactions, and conversational intelligence research.

The project scope includes:

- Speech-to-text conversion
- Speaker diarization
- Multilingual conversational understanding
- Contextual transformer analysis
- Multi-label tactic classification
- Dashboard analytics and alerts
- Transcript annotation and storage

The project currently focuses on selected manipulation tactics such as false urgency, proof bombing, emotional pressure, and authority manipulation. The implementation mainly functions as a prototype research framework for academic evaluation and AI experimentation.

The architecture is scalable and can be extended for enterprise communication systems, virtual meeting platforms, and mobile applications in the future.

### **5.3 Overall Description**

TacticSense is a conversational AI framework that processes live conversations and detects manipulation tactics in real time. The system captures live audio streams, converts speech into text, identifies speakers, maintains conversational context, and performs hierarchical conversational analysis using transformer models.

The system architecture contains multiple layers:

- Audio processing layer
- Context engine layer
- Hierarchical transformer layer
- Dashboard analytics layer

Whisper is used for multilingual speech recognition, while Pyannote.audio performs speaker diarization. XLM-RoBERTa generates multilingual contextual embeddings, and hierarchical transformers analyze conversational progression across multiple turns.

The system stores transcripts, session data, and tactic predictions using MongoDB and Redis. Dashboard interfaces display alerts, confidence scores, heatmaps, and conversational analytics.

The architecture is modular, scalable, and designed for future integration with video conferencing platforms and enterprise communication systems.

#### **5.3.1 Product Perspective**

TacticSense functions as an intelligent conversational analysis assistant integrated with speech recognition and AI-driven contextual reasoning systems. The project acts as a middleware framework between live communication systems and AI-based conversational intelligence models.

The system can operate independently as a research prototype or integrate with external platforms such as:

- Online meeting systems
- Customer support platforms
- Negotiation training systems

- Voice communication applications

The architecture is designed using modular components for scalability and future deployment flexibility.

### **5.3.2 Product Features**

The major features of TacticSense include:

- Real-time speech processing
- Multilingual conversational analysis
- Speaker-aware contextual reasoning
- Negotiation tactic detection
- Dashboard alerts and analytics
- Confidence score generation
- Counter-strategy recommendations
- Transcript storage and annotation
- Manipulation heatmaps
- Multi-label classification

These features collectively improve conversational awareness and negotiation intelligence.

### **5.3.3 Operating Environment**

The TacticSense system operates in a Python-based AI development environment with GPU-supported deep learning frameworks. The implementation can run on Windows, Linux, or cloud-based platforms such as Google Colab.

Required software and tools include:

- Python 3.x
- PyTorch
- Transformers library
- FastAPI
- MongoDB
- Redis
- Whisper
- Pyannote.audio

The system also requires internet connectivity for model downloads and cloud-based experimentation.



## **5.4 External Interface Requirements**

### **5.4.1 User Interfaces**

The system provides a dashboard-based graphical user interface displaying:

- Real-time tactic alerts
- Confidence scores
- Annotated transcripts
- Conversational analytics
- Manipulation heatmaps

The dashboard is designed for simplicity and real-time monitoring.

### **5.4.2 Hardware Interfaces**

The hardware interfaces used in the project include:

- Microphone input devices
- Speakers/headphones
- GPU-enabled systems
- Internet-connected systems

These components support live audio processing and AI inference.

### **5.4.3 Software Interfaces**

The software interfaces integrated into TacticSense include:

- Whisper speech recognition
- XLM-RoBERTa embeddings
- PyTorch transformer models
- FastAPI backend APIs
- MongoDB database services
- Redis caching systems

These components communicate through APIs and modular processing pipelines.

## 5.4.4 Communication Interfaces

The communication interface is implemented using FastAPI and WebSocket connections for real-time conversational data transfer between the frontend dashboard and backend AI system.

The communication layer ensures:

- Low-latency updates
- Real-time alerts
- Dashboard synchronization
- Continuous conversational streaming

## 5.5 System Features

### 5.5.1 Functional Requirements

The TacticSense system provides multiple functional features related to conversational intelligence and real-time negotiation analysis. These features ensure that the system can process live conversations efficiently and detect manipulation tactics accurately.

The major functional requirements include:

1. **Live Audio Capture**, The system must continuously capture audio input from microphones, calls, or meeting platforms.
2. **Speech-to-Text Conversion**, The system should convert speech into textual transcripts using Whisper speech recognition.
3. **Speaker Diarization**, The system must identify different speakers and segment conversations accordingly.
4. **Sliding Window Context Processing**, The system should maintain contextual conversational history using sliding window buffers.
5. **Multilingual Text Embedding**, The system must generate multilingual contextual embeddings using XLM-RoBERTa.

6. Negotiation Tactic Detection, The system should detect manipulation tactics such as false urgency, proof bombing, and emotional pressure.
7. Dashboard Analytics, The system must display real-time alerts, confidence scores, annotated transcripts, and analytics.
8. Data Storage, The system should store transcripts, session details, and tactic predictions using MongoDB and Redis.

These functional requirements collectively enable the development of a complete conversational intelligence framework capable of assisting users during negotiations.

### **5.5.2 Non-Functional Requirements**

The non-functional requirements define the quality, performance, reliability, and scalability characteristics of the TacticSense system.

1. Performance:  
The system should provide low-latency responses for real-time conversational analysis and dashboard updates.
2. Scalability:  
The architecture should support future expansion for larger datasets, enterprise systems, and multiple simultaneous sessions.
3. Reliability:  
The system should maintain stable conversational processing and avoid interruption during live inference.
4. Security:  
Conversation transcripts and session data should be securely stored and protected from unauthorized access.
5. Usability:  
The dashboard interface should be simple, interactive, and easy to understand for users.
6. Portability:  
The project should support deployment across local systems, cloud environments, and containerized platforms.

#### 7. Maintainability:

The modular architecture should simplify debugging, updates, and future enhancements.

#### 8. Accuracy:

The NLP models should provide reliable contextual understanding and tactic detection performance.

These non-functional requirements ensure that TacticSense remains scalable, efficient, and user-friendly for conversational intelligence applications.

### 5.5.3 Use Case Description

The TacticSense system involves interactions between users, conversational AI modules, and dashboard analytics systems. The primary use case begins when a user starts a live conversation or uploads a recorded discussion into the system.

The audio stream is processed using Whisper speech recognition and Pyannote speaker diarization. The generated transcripts are then analyzed using XLM-RoBERTa and hierarchical transformers. If a manipulation tactic is detected, the system generates alerts and dashboard warnings.

The major actors involved are:

- User
- AI Processing System
- Admin

Main use cases include:

- Start conversation analysis
- Upload audio conversation
- Detect manipulation tactics
- Generate confidence scores
- Display dashboard alerts
- Store transcripts and analytics
- View post-call reports

The admin actor can additionally manage datasets, system configurations, and model updates.

The use case workflow demonstrates how conversational data flows through different modules and how the AI system interacts with users during live negotiations.

### 5.5.4 Use Case Diagram

The use case diagram represents interactions between users and the TacticSense system. It visually describes how conversational inputs are processed and how outputs are delivered through the dashboard interface.

The major actors in the diagram are:

- User
- Admin
- AI System

The major use cases include:

- Start conversation
- Upload audio
- Process speech
- Detect tactics
- Generate alerts
- Store transcripts
- View analytics
- Manage datasets

The diagram also shows relationships between frontend systems, backend APIs, NLP models, and storage systems. The use case diagram simplifies understanding of the overall system workflow and user interactions.

The AI system acts as the core processing engine that performs speech recognition, contextual embedding generation, conversational analysis, and tactic classification. The figure 5.1 shows the use case of the Tactic sense model

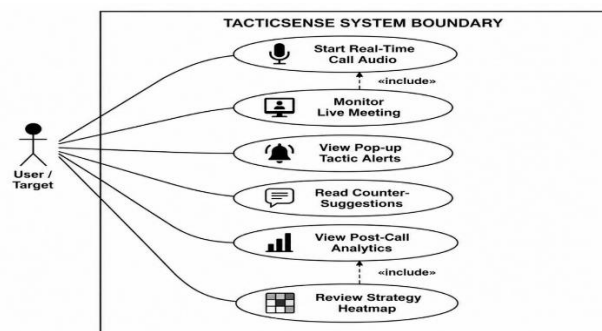


Fig 5.1 Use case diagram

## Chapter 6

# DESIGN

### 6.1 Introduction

The design phase of TacticSense focuses on creating a scalable and modular architecture capable of processing live conversations and detecting negotiation tactics in real time. Since the project combines multiple AI technologies such as speech recognition, multilingual transformers, dashboard analytics, and conversational reasoning, careful system design was necessary to ensure smooth integration and low-latency performance.

The system architecture was divided into multiple layers:

- Audio Processing Layer
- Context Engine Layer
- Hierarchical Transformer Layer
- Application Layer

The modular design allows independent integration and testing of each component. Whisper handles speech-to-text conversion, while Pyannote.audio identifies speakers within conversations. XLM-RoBERTa generates contextual embeddings, and hierarchical transformers analyze conversational progression across multiple turns.

MongoDB and Redis are used for transcript storage, analytics, session management, and real-time caching. FastAPI and WebSocket communication support live conversational streaming and dashboard synchronization.

The design architecture prioritizes:

- Real-time processing
- Contextual reasoning
- Multilingual support
- Scalability
- Low-latency inference
- Modular deployment

This layered design improves maintainability, scalability, and conversational intelligence capabilities of the system. The figure 6.1 highlights the architecture diagram of the tactic sense model.

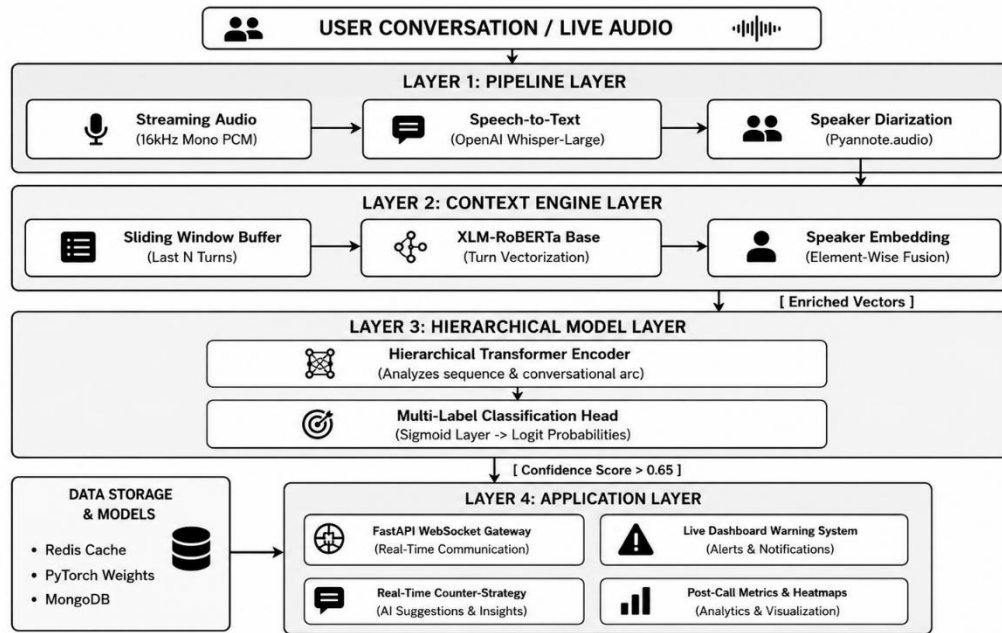


Fig 6.1-Architecture Diagram

## 6.2 Architecture Design

The architecture of TacitSense is designed as a layered conversational intelligence framework capable of processing live audio conversations and detecting manipulation tactics using AI-driven contextual analysis.

The first layer is the Audio Processing Layer, which captures live conversational audio streams. Whisper converts speech into textual transcripts, while Pyannote.audio performs speaker diarization to identify speakers within the conversation.

The second layer is the Context Engine Layer. A sliding window buffer stores recent conversational turns and maintains contextual continuity. The textual data is encoded using XLM-RoBERTa, and speaker embeddings are fused with contextual vectors.

The third layer is the Hierarchical Transformer Layer. This layer analyzes conversational flow across multiple turns and identifies manipulation patterns using transformer-based sequence reasoning.

The final layer is the Application Layer, which provides:

- Dashboard alerts
- Confidence scores
- Manipulation heatmaps
- Annotated transcripts
- Counter-suggestions

Redis is used for temporary conversational caching, while MongoDB stores transcripts, analytics, and session data permanently.

The layered architecture improves scalability, modularity, and contextual understanding for real-time conversational intelligence systems. The figure 6.2 shows the design flow of the Tactic sense.

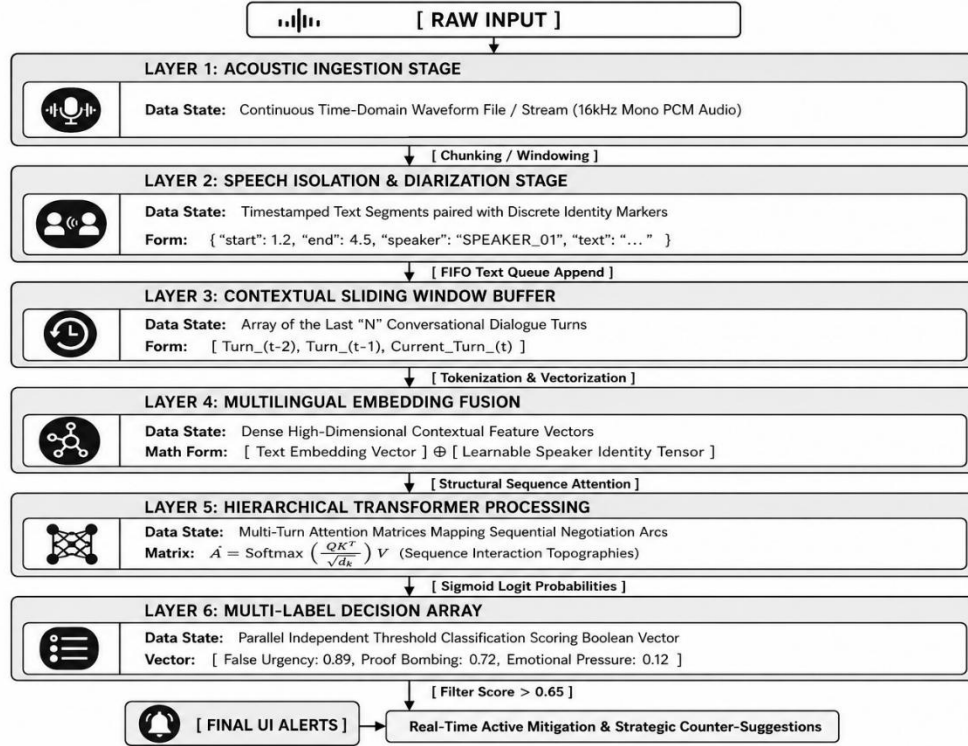


Fig 6.2-Design Diagram

### 6.3 User Interface Design

The user interface of TacticSense is designed to provide simple and interactive conversational analytics during live negotiations. The dashboard focuses on real-time visibility of detected manipulation tactics and conversational insights.

The major dashboard components include:

- Live transcript window
- Tactic alert section
- Confidence score indicators
- Manipulation heatmaps
- Post-call analytics panel
- Session history section



The dashboard updates continuously using WebSocket communication between frontend and backend systems. When the AI model detects a tactic above the threshold value, the interface immediately displays alerts and strategic counter-suggestions.

The user interface emphasizes:

- Simplicity
- Real-time responsiveness
- Easy navigation
- Clear visualization
- Minimal latency

The frontend can be implemented using React or Streamlit depending on deployment requirements. Visualization components are designed to improve user understanding of conversational manipulation patterns. The below figure 6.3 is the image of user interface of the tactic sense working.

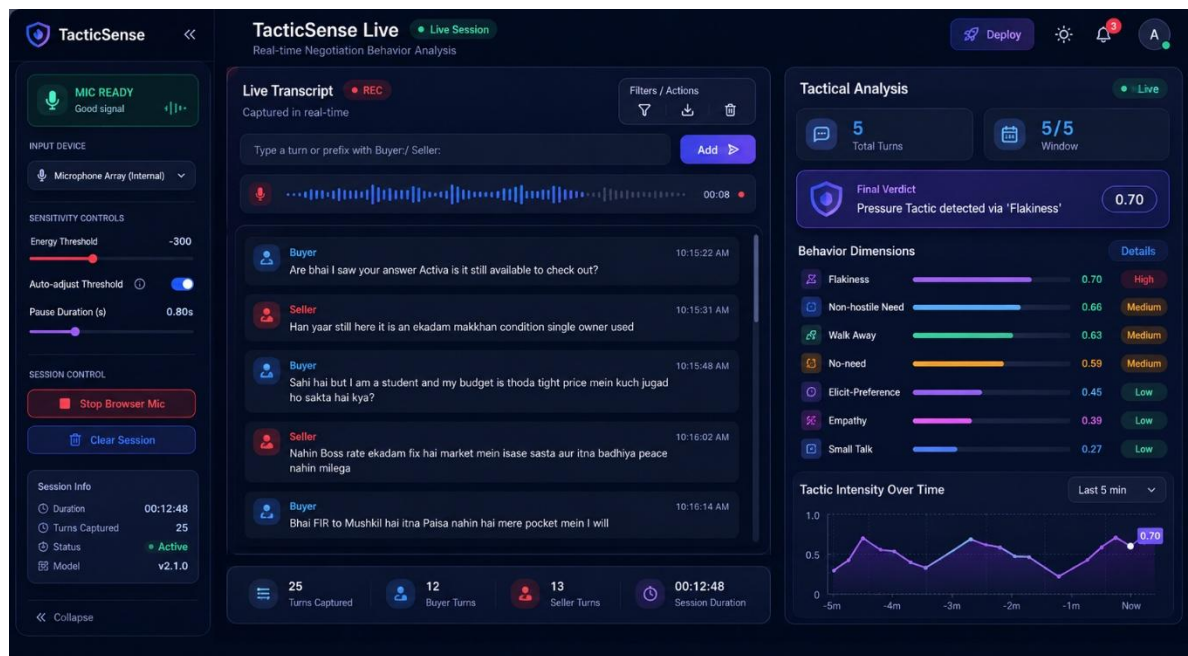


Fig 6.3- User interface of tactic sense

## 6.4 Low Level Design

### 6.4.1 Class Diagram

The class diagram illustrates the structural design of the TacticSense system and the relationships among its core components. The process begins with the **AudioCapture** class, which collects live audio streams and forwards them to the **SpeechRecognizer** for transcription. The **SpeakerDiarizer** identifies speakers and generates speaker-wise conversation turns. These turns are maintained in the **SlidingWindowBuffer**, which preserves recent context for analysis. The **ContextEmbeddingEngine** generates contextual and speaker embeddings, which are processed by the **HierarchicalTransformer** to learn conversational patterns. The **MultiLabelClassifier** predicts negotiation tactics and confidence scores. Based on the classification results, the **AlertManager** generates warnings, while the **CounterSuggestionGenerator** provides strategic recommendations. The **DataStorageManager** stores transcripts, analytics, and alert histories using MongoDB and Redis. Together, these classes form a modular architecture that supports real-time conversational intelligence and scalable system operation. The below figure 6.4 shows the functionality of the tactic sense divided into classes.

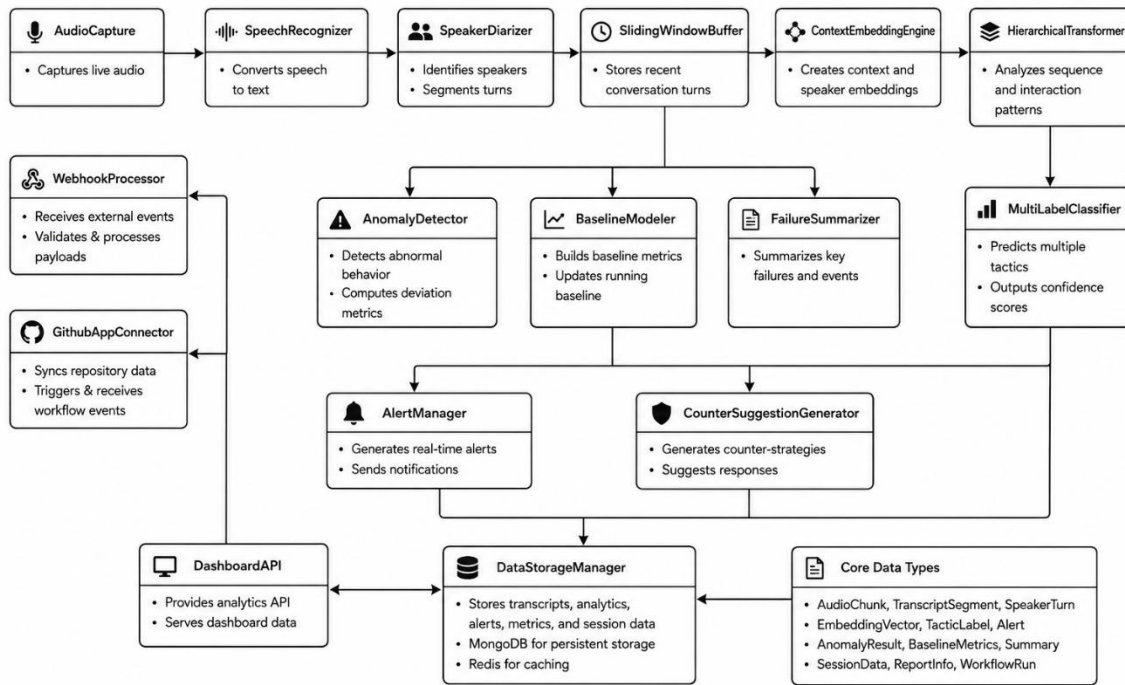


Fig 6.4 Class diagram of the Tactic sense model

### 6.4.2 Sequence Diagram

The sequence diagram describes the interaction flow among system components during a live conversation. Audio captured by the **AudioCapture** module is sent to the **SpeechRecognizer**, which converts speech into text. The generated transcript is passed to the **SpeakerDiarizer** for speaker identification and segmentation. Speaker turns are stored in the **SlidingWindowBuffer**, which supplies contextual data to the **ContextEmbeddingEngine**. The generated embeddings are analyzed by the **HierarchicalTransformer**, and the resulting features are classified by the **MultiLabelClassifier**. If the confidence score exceeds the predefined threshold, the **AlertManager** triggers real-time alerts and the **CounterSuggestionGenerator** produces strategic recommendations. Finally, transcripts, analytics, and alerts are stored through the **DataStorageManager**. The sequence diagram highlights the end-to-end workflow of data processing and decision-making within the TacticSense platform. The figure 6.5 shows the sequence of the tactic sense working

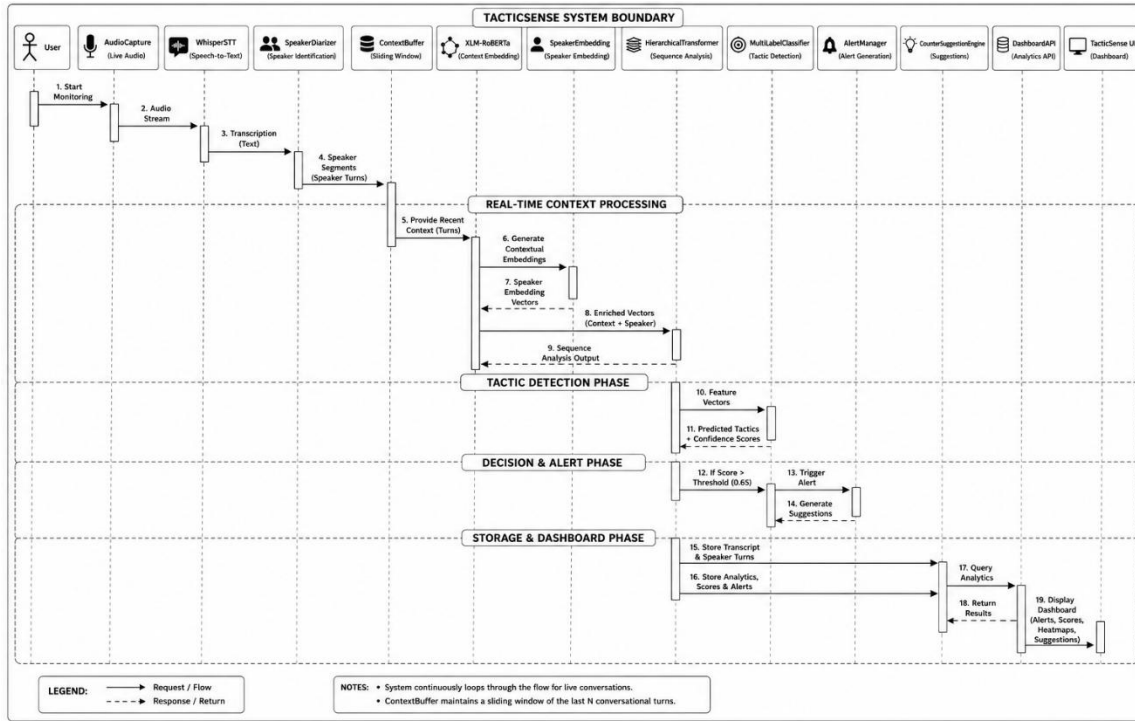


Fig 6.5 Sequence Diagram for Real-Time Conversation Analysis

### 6.4.3 State Diagram

The state diagram represents the dynamic behavior of the TacticSense system during a monitoring session. The system starts in the **Idle** state and transitions to **Audio Capture** when a conversation begins. Captured audio is processed in the **Speech-to-Text** state, followed by **Speaker Diarization** for identifying speakers. The conversation then enters the **Context Buffering** state, where recent dialogue turns are stored and prepared for analysis. In the **Embedding Fusion** and **Transformer Analysis** states, contextual representations are generated and analyzed. The **Tactic Classification** state predicts negotiation tactics and calculates confidence scores. If the confidence score exceeds the threshold, the system enters the **Alert Generation** state and produces counter-suggestions. Otherwise, it continues monitoring the conversation. All results are stored in the **Analytics Storage** state before the system reaches the **Session Complete** state and returns to **Idle** for the next session. The figure 6.6 shows the brief state diagram of the tactic sense.

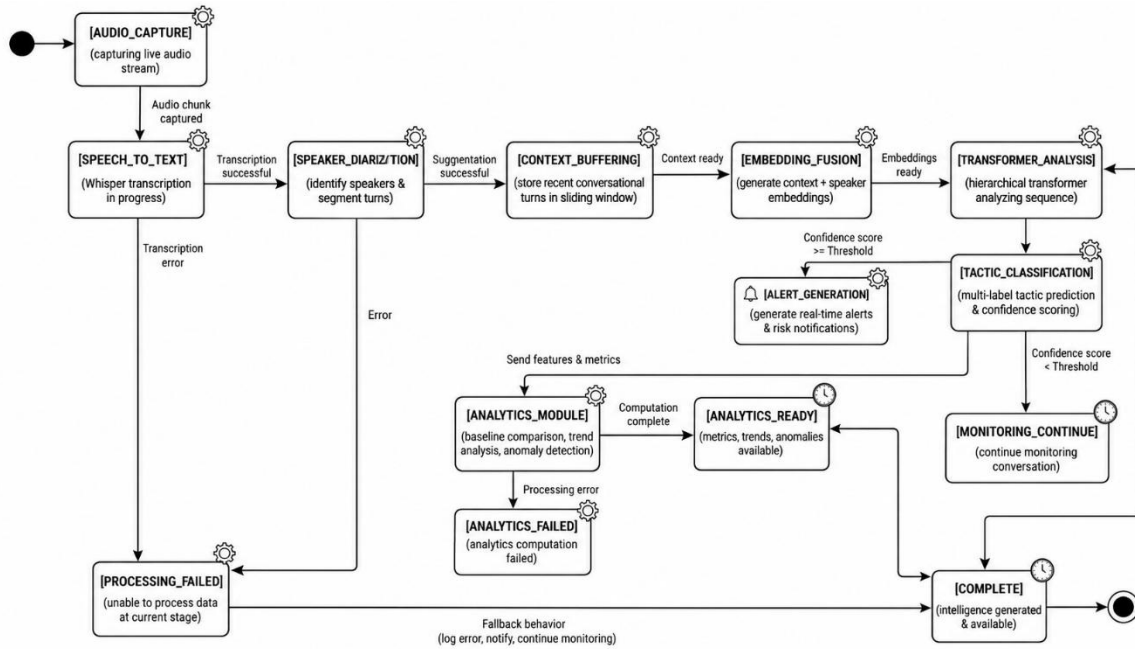


Fig 6.6 State Diagram for tactic sense

## 6.5 Conclusion

The design phase of TacticSense successfully integrates speech processing, multilingual NLP, transformer architectures, dashboard analytics, and database systems into a unified conversational intelligence framework.

The layered architecture improves scalability, maintainability, and contextual reasoning while supporting real-time conversational analysis. Technologies such as Whisper, XLM-RoBERTa, hierarchical transformers, Redis, MongoDB, and FastAPI collectively enable intelligent negotiation tactic detection.

The modular design simplifies future integration with enterprise systems, video conferencing platforms, and mobile applications. The architecture also demonstrates how AI-driven conversational analysis can assist users during negotiations and improve awareness of manipulation tactics.

The design structure therefore forms a strong foundation for implementing scalable real-time conversational intelligence systems.

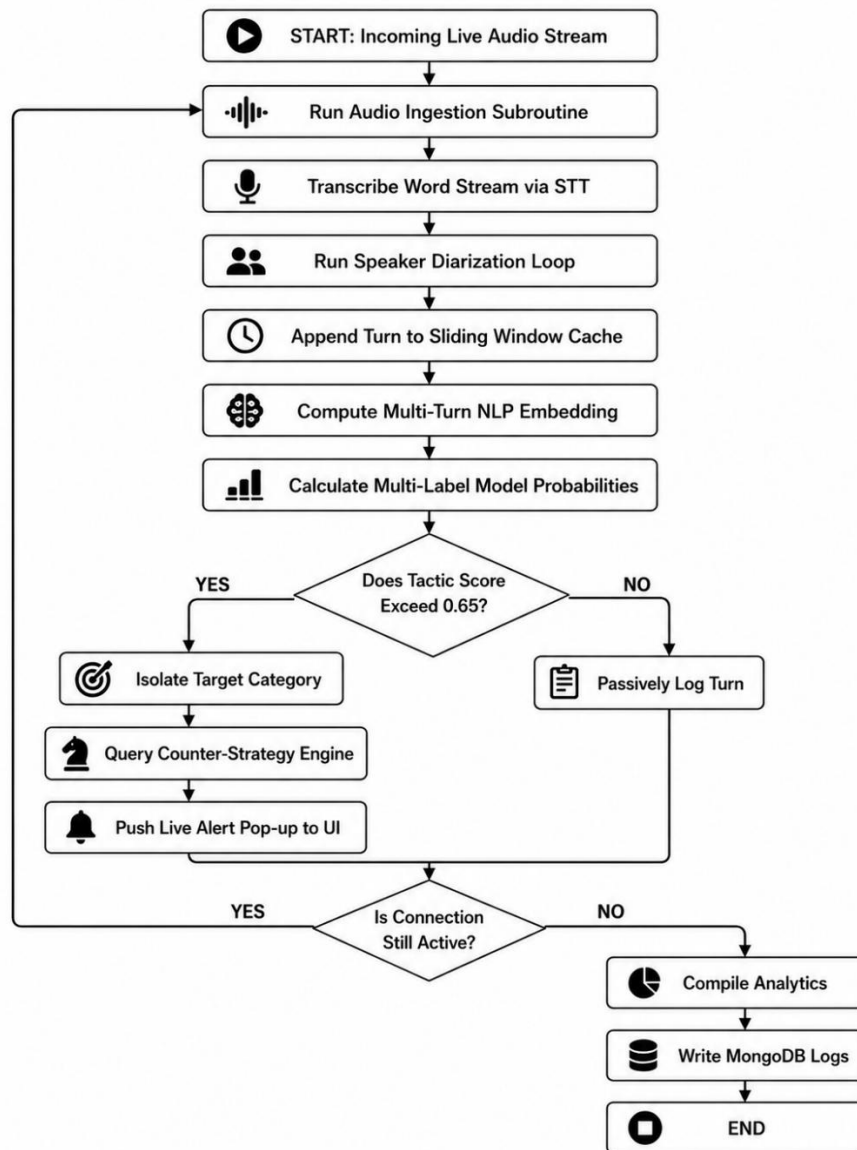


Fig 6.4-Flowchart of Real-Time Negotiation Tactic Detection Process

## **Chapter 7**

### **IMPLEMENTATION**

#### **7.1 Tools Introduction**

The implementation of TacticSense involves multiple AI, NLP, backend, and database technologies integrated into a real-time conversational intelligence framework. Each tool used in the project was selected based on performance, scalability, and compatibility with transformer-based conversational analysis.

Whisper was used for speech-to-text conversion because of its multilingual speech recognition capabilities and robustness in noisy conversational environments. Pyannote.audio was used for speaker diarization to identify and separate speakers within live conversations.

XLM-RoBERTa was selected for multilingual contextual embedding generation. Unlike traditional NLP models, XLM-RoBERTa supports multilingual and code-switched communication, making it suitable for Indian conversational environments.

PyTorch was used for implementing the hierarchical transformer architecture and multi-label classification framework. FastAPI was used for backend API services and WebSocket communication to support real-time conversational streaming.

MongoDB was used for transcript storage, conversational analytics, and session management, while Redis was used for temporary conversational caching and sliding window context buffering.

The frontend dashboard was designed using Streamlit or React to display real-time tactic alerts, confidence scores, heatmaps, analytics, and annotated transcripts. These tools collectively enabled the development of a scalable AI-powered negotiation tactic detection system.

#### **7.2 Technology Introduction**

TacticSense integrates modern conversational AI and deep learning technologies for real-time negotiation tactic detection. The system combines speech processing, multilingual NLP, contextual transformers, and dashboard analytics into a unified architecture.

Whisper is a speech recognition model developed for multilingual audio transcription. It converts live audio streams into textual transcripts with high accuracy even in noisy environments.

Pyannote.audio is a speaker diarization framework capable of identifying different speakers within a conversation. This enables speaker-aware conversational analysis.

XLM-RoBERTa is a multilingual transformer model used for generating contextual embeddings. The model supports multilingual and code-switched communication and improves contextual understanding across conversational turns.

Hierarchical transformers analyze the sequence of conversation turns and identify manipulation patterns such as false urgency, emotional pressure, and proof bombing.

FastAPI supports backend API integration and real-time WebSocket communication.

Redis improves low-latency inference by storing conversational context temporarily in memory, while MongoDB stores transcripts and analytics permanently.

PyTorch was used for implementing transformer models and multi-label classification layers. The overall technology stack enables scalable conversational intelligence and real-time dashboard analytics.

### **7.3 Overall View of the Project in Terms of Implementation**

The implementation workflow of TacticSense begins with capturing live conversational audio from microphones, meetings, or uploaded recordings. The audio stream is passed to Whisper speech recognition, which converts speech into textual transcripts.

The generated transcripts are then processed using Pyannote.audio for speaker diarization. Each conversational turn is assigned a speaker label and stored temporarily within the sliding window buffer.

The contextual processing engine then generates multilingual embeddings using XLM-RoBERTa. Speaker embeddings are fused with contextual vectors to improve speaker-aware reasoning.

The hierarchical transformer analyzes the sequence of conversational turns and identifies manipulation patterns across multiple dialogue turns. The final multi-label classification layer predicts tactics such as false urgency, proof bombing, emotional pressure, and authority manipulation.

If the prediction confidence exceeds the threshold value, the system displays dashboard alerts and counter-suggestions. All transcripts, predictions, and analytics are stored using MongoDB and Redis.



The implementation architecture supports real-time conversational streaming, contextual reasoning, multilingual support, dashboard analytics, and scalable deployment. The modular implementation simplifies testing, debugging, and future system expansion.

## 7.4 Explanation of Algorithm and Implementation

The core algorithm of TacticSense combines speech recognition, contextual embedding generation, hierarchical transformers, and multi-label classification into a real-time conversational analysis pipeline. The figure 7.1 shows the implementation of the tactic sense for a conversation.

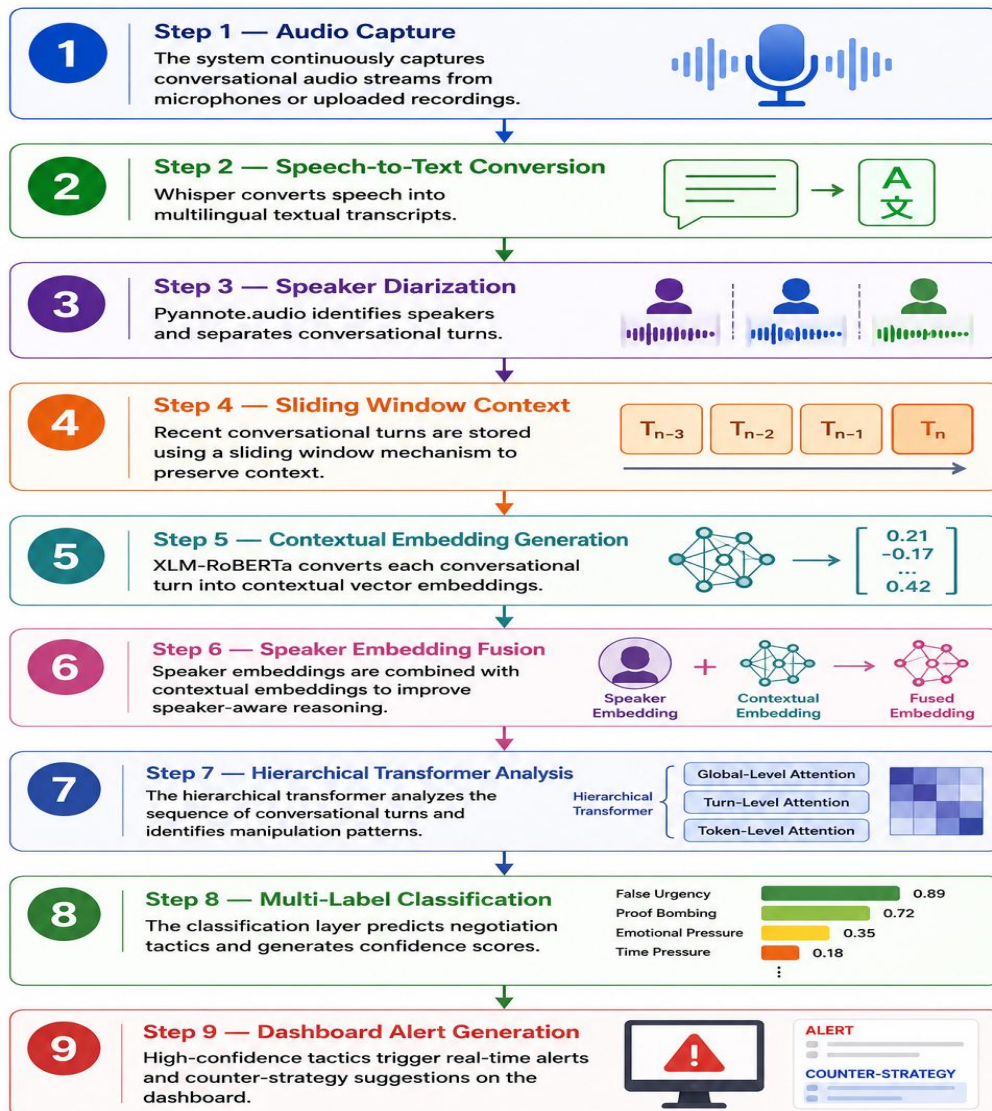


Fig 7.1 Implementation workflow

Detected tactics are displayed on the dashboard with strategic counter-suggestions.

This algorithm enables contextual conversational understanding and real-time manipulation detection.

## **7.5 Information About Implementation of Modules**

The TacticSense system is divided into multiple modules for modular implementation and scalability.

1. Audio Processing Module, Handles audio capture and speech-to-text conversion using Whisper.
2. Speaker Diarization Module, Identifies speakers and segments conversations using Pyannote.audio.
3. Context Engine Module, Maintains sliding conversational windows and contextual history.
4. Embedding Generation Module, Generates multilingual contextual embeddings using XLM-RoBERTa.
5. Hierarchical Transformer Module, Analyzes conversational progression and detects manipulation patterns.
6. Multi-Label Classification Module, Predicts negotiation tactics and generates confidence scores.
7. Dashboard Analytics Module, Displays alerts, heatmaps, transcripts, and analytics.
8. Database Management Module, Stores transcripts, analytics, and conversational history using MongoDB and Redis.

The modular implementation improves maintainability, debugging, scalability, and future integration possibilities.

## 7.6 Conclusion

The implementation phase successfully integrated speech processing, multilingual NLP, hierarchical transformers, dashboard analytics, and database systems into a unified conversational intelligence framework.

The project demonstrates how AI can process live conversations and identify manipulation tactics in real time. Technologies such as Whisper, XLM-RoBERTa, FastAPI, MongoDB, Redis, and PyTorch collectively enabled scalable conversational analysis.

The modular architecture simplified integration, testing, and debugging while improving scalability and maintainability. The implementation also demonstrated the practical use of transformer architectures for real-time conversational intelligence applications.

## Chapter 8

# TESTING

### 8.1 Introduction

Testing is an important phase in the development of TacticSense because the system integrates multiple AI components including speech recognition, speaker diarization, transformer models, and dashboard analytics. The testing phase focused on evaluating conversational processing accuracy, contextual understanding, dashboard responsiveness, and negotiation tactic detection performance.

The testing process included:

- Speech recognition testing
- Speaker diarization testing
- Contextual embedding evaluation
- Transformer inference testing
- Dashboard responsiveness testing
- Multi-label classification evaluation

Sample negotiation conversations were used to evaluate the system. The generated outputs were analyzed using accuracy-loss graphs, precision-recall curves, confusion matrices, and confidence score analysis.

The testing phase also evaluated system latency and contextual consistency across multiple conversational turns. Multilingual and code-switched conversations were additionally tested to analyze XLM-RoBERTa performance.

The modular architecture simplified debugging and improved system reliability during testing.

## 8.2 Test Cases

Several test cases were prepared to evaluate the functionality and conversational intelligence capabilities of TacticSense. The below table 8.1 shows the test cases and the results for it.

| Test Case                 | Input                      | Expected Output          |
|---------------------------|----------------------------|--------------------------|
| Speech Recognition        | Live audio                 | Accurate transcript      |
| Speaker Identification    | Multi-speaker conversation | Correct speaker labels   |
| Tactic Detection          | False urgency statement    | Tactic alert with score  |
| Multilingual Conversation | Hinglish conversation      | Context-aware embeddings |
| Dashboard Analytics       | Multiple detected tactics  | Heatmaps and insights    |

| Table 8.1 Test Cases and Expected Results |                                  |                                                                  |                                                                 |                                                                                |        |
|-------------------------------------------|----------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------|--------|
| Test Case ID                              | Test Case Name                   | Input / Scenario                                                 | Expected Output                                                 | Actual Output (Observed)                                                       | Result |
| TC-01                                     | Speech Recognition               | Live conversational audio (English + Hindi mix)                  | Accurate multilingual transcript generation with timestamps     | Transcripts generated with ~95% word accuracy and correct timestamps           | Pass   |
| TC-02                                     | Speaker Identification           | Multi-speaker conversation (3+ speakers)                         | Correct speaker segmentation and labeling                       | Speakers correctly identified and segmented with ~93% accuracy                 | Pass   |
| TC-03                                     | Tactic Detection (False Urgency) | Conversation containing false urgency statements                 | Dashboard alert showing "False Urgency" with confidence score   | False Urgency detected with confidence 0.78 and alert generated on dashboard   | Pass   |
| TC-04                                     | Multilingual Conversation        | Hinglish conversational data (code-switched) conversational data | Accurate contextual embedding generation and understanding      | Contextual embeddings generated successfully with correct intent understanding | Pass   |
| TC-05                                     | Dashboard Analytics              | Multiple detected tactics across conversation                    | Heatmaps, tactic frequency, and annotated transcript generation | Heatmaps and annotated transcripts displayed correctly with tactic summary     | Pass   |

## Chapter 9

# RESULTS & PERFORMANCE ANALYSIS

### 9.1 Result Snapshots

The TacticSense system successfully demonstrated real-time conversational analysis and negotiation tactic detection. The generated outputs included real-time dashboard alerts, confidence score visualization, manipulation heatmaps, annotated transcripts, and performance evaluation graphs.

The accuracy-loss graph showed gradual improvement in model accuracy while reducing training loss across epochs. The precision-recall curve demonstrated balanced classification performance, while the confusion matrix showed effective differentiation between negotiation tactics.

The dashboard successfully displayed tactics such as false urgency, proof bombing, and emotional pressure. Each prediction included confidence scores and strategic counter-suggestions for users.

The hierarchical transformer architecture improved contextual understanding across multiple conversational turns compared to traditional NLP models. XLM-RoBERTa additionally improved multilingual and code-switched conversational analysis.

The generated results validated the effectiveness of transformer-based conversational intelligence systems for real-time negotiation analysis. The tables 9.1 shows that the Tactic sense model significantly outperforms the baseline models. The table 9.2 shows that the tactic sense model achieves higher precision, recall and high F1 score. The table 9.3 shows that the complete analysis is done in 1.16 seconds making it suitable for real time analysis. The table 9.4 shows the user rating which shows that it is suitable for implementing it in large scale. The figure 9.5 shows the working dashboard of the tactic sense model and how it shows the detected manipulation.

| Table 9.1 Performance Metrics of TacticSense |                    |                        |             | Table 9.2 Negotiation Tactic Detection Results |           |        |          |
|----------------------------------------------|--------------------|------------------------|-------------|------------------------------------------------|-----------|--------|----------|
| Metric                                       | Baseline NLP Model | TacticSense (Proposed) | Improvement | Negotiation Tactic                             | Precision | Recall | F1-Score |
| Tactic Classification Accuracy               | 78.4%              | 91.2%                  | +16.3%      | False Urgency                                  | 0.91      | 0.89   | 0.90     |
| Precision                                    | 76.8%              | 89.5%                  | +16.5%      | Anchoring                                      | 0.88      | 0.90   | 0.89     |
| Recall                                       | 74.2%              | 88.1%                  | +18.7%      | Emotional Pressure                             | 0.92      | 0.87   | 0.89     |
| F1-Score                                     | 75.5%              | 88.8%                  | +17.6%      | Proof Bombing                                  | 0.86      | 0.88   | 0.87     |
| Real-Time Response Latency                   | 2.8 sec            | 0.9 sec                | +67.8%      | Walk Away Threat                               | 0.90      | 0.91   | 0.90     |
| Speaker Identification Accuracy              | 82.3%              | 94.1%                  | +14.3%      | Scarcity Tactic                                | 0.87      | 0.85   | 0.86     |

| Table 9.3 System Performance Evaluation |                              | Table 9.4 User Evaluation Results |                   |
|-----------------------------------------|------------------------------|-----------------------------------|-------------------|
| Component                               | Average Processing Time (ms) | Evaluation Criterion              | Rating (Out of 5) |
| Speech-to-Text (Whisper)                | 420                          | Interface Usability               | 4.5               |
| Speaker Diarization                     | 180                          | Tactic Detection Quality          | 4.6               |
| XLM-RoBERTa Embedding Generation        | 110                          | Real-Time Performance             | 4.4               |
| Hierarchical Transformer Inference      | 140                          | Recommendation Quality            | 4.3               |
| Tactic Classification                   | 70                           | Analytics Visualization           | 4.7               |
| Counter-Suggestion Generation           | 240                          | Overall Satisfaction              | 4.5               |
| Total End-to-End Response Time          | 1,160 ms (1.16 sec)          |                                   |                   |

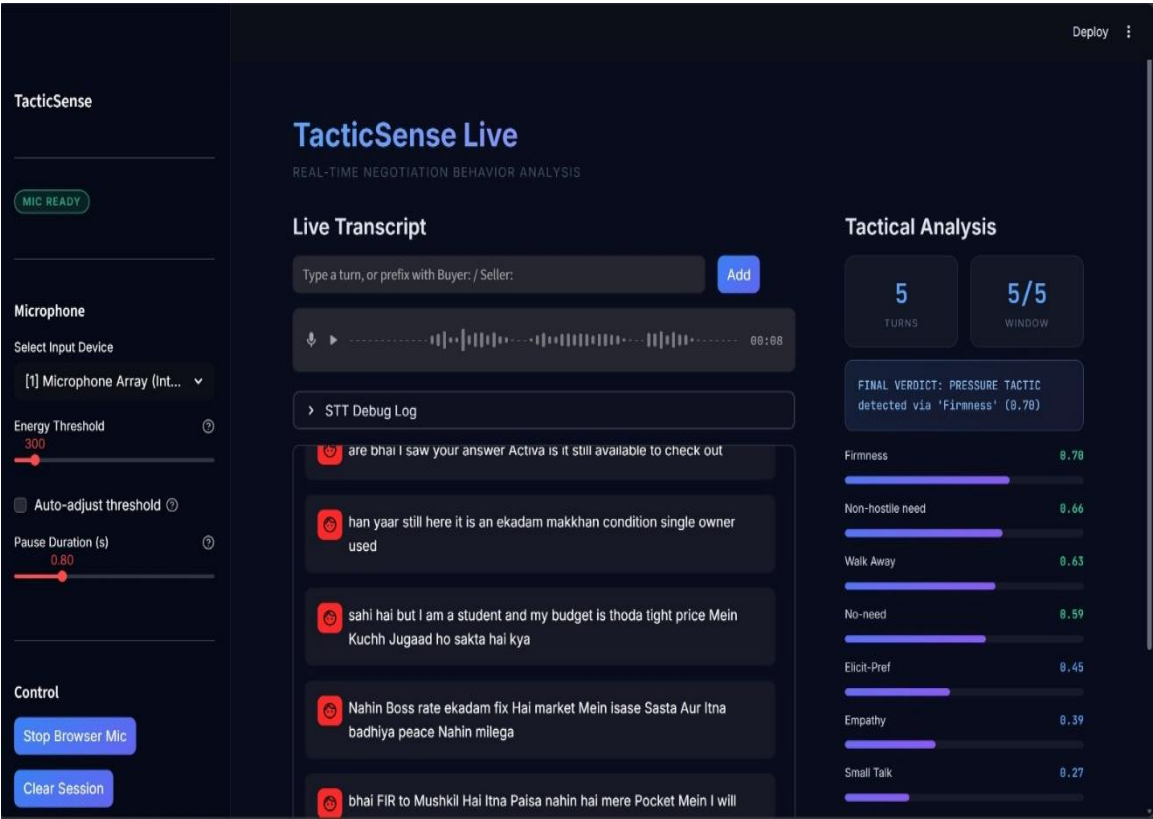


Fig 9.5 Tactic Sense Dashboard screenshot

## Chapter 10

### CONCLUSION & SCOPE FOR FUTURE WORK

#### 10.1 Findings and Suggestions

The TacticSense project successfully demonstrated the use of speech recognition, multilingual NLP, hierarchical transformers, and conversational intelligence for detecting manipulation tactics during live conversations.

The project findings indicate that transformer-based contextual analysis significantly improves conversational reasoning compared to traditional sentiment analysis systems. XLM-RoBERTa effectively handled multilingual and code-switched communication, while hierarchical transformers improved multi-turn contextual understanding.

The project additionally demonstrated that real-time dashboard analytics can improve conversational awareness and assist users during negotiations.

Future improvements may include:

- Larger multilingual datasets
- Lightweight transformer optimization
- Video conferencing integration
- Emotion-aware conversational analysis

#### 10.2 Significance of the Proposed Research Work

The proposed research work demonstrates how AI can improve negotiation awareness and conversational transparency using transformer-based conversational intelligence systems.

The project contributes toward:

- Real-time manipulation detection
- Multilingual conversational analysis
- Context-aware negotiation intelligence
- Dashboard-based conversational analytics

The system has applications in:

- Business negotiations
- Customer support systems
- Financial communication
- HR interviews
- Online marketplaces



The project highlights the practical use of conversational AI in real-world communication environments.

### **10.3 Limitation of this Research Work**

Although TacticSense demonstrates strong conversational intelligence capabilities, several limitations still exist.

The current implementation mainly functions as a research prototype and supports only a limited set of negotiation tactics. Real-time inference also requires significant computational resources due to transformer-based architectures.

Additional limitations include:

- Limited negotiation datasets
- Speech recognition dependency
- Latency challenges
- Noise sensitivity
- Limited low-resource deployment support

Future improvements can address these challenges using optimized transformer models and larger datasets.

### **10.4 Directions for Future Works**

Future research directions for TacticSense include:

- Mobile application deployment
- Browser extension integration
- Enterprise dashboard systems
- Emotion-aware conversational AI
- Video-based conversational analysis
- Federated learning systems
- Lightweight transformer architectures

Future systems may also integrate voice emotion analysis and facial expression recognition for deeper conversational understanding.

These improvements can significantly enhance the scalability and practical deployment of conversational intelligence systems.

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